

AN INTEGRATED AI, VR, AND AR BASED WEDDING PLANNING SYSTEM: AR Integration Module

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Dissertation submitted in partial fulfillment of the requirements for the Bachelor of
Science (Hons) in Information Technology Specializing in Interactive Media

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April 2026

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ABSTRACT

The contemporary wedding planning process is fraught with logistical and financial uncertainties, largely due to the disconnect between imagination, physical venue constraints, and budget management. Traditional planning approaches rely heavily on static images, spreadsheets, and fragmented tools, which severely limit users' ability to visually and financially evaluate their decisions. This research addresses these spatial and cognitive limitations through the development of a mobile Augmented Reality (AR) decor customization and budget tracking system. Functioning as a core pillar of a larger integrated Artificial Intelligence (AI), VR, and AR wedding planning platform, this mobile system shifts the planning paradigm from imagination-based to highly interactive spatial decision-making directly on-site. Developed natively for the iOS ecosystem utilizing Swift, ARKit, and RealityKit, the application facilitates markerless environmental tracking and high-fidelity 3D spatial visualization within physical venues.

A primary functional contribution of this system is its unique architectural integration of real-time financial tracking bound directly to spatial actions within the AR interface. As users dynamically place and manipulate 3D cultural assets, such as the traditional Poruwa or banquet seating arrays, the system utilizes rule-based logic to instantly update local currency (LKR) budgets and recalculate maximum venue capacities. To bridge the gap between virtual spatial planning and physical execution, a Gemini AI-powered vendor suggestion engine was integrated. This module autonomously formats the user's spatial and financial state into JSON schemas, filtering localized catalogs of Sri Lankan vendors based on strict financial constraints. Technical evaluations demonstrate high spatial accuracy via plane detection, maintaining a fluid 60 FPS during standard object rendering and highly optimized API parsing speeds (under 250ms). Comprehensive user testing with 20 participants including professional planners, venue managers, and engaged couples confirmed significantly higher spatial understanding, logistical confidence, and conceptual retention, definitively validating the commercial utility of immersive AR planning tools in the event industry.

Keywords: Augmented Reality, Wedding Planning, ARKit, Budget Tracking, Swift Development, Generative AI Integration, Spatial UI.

ACKNOWLEDGEMENTS

I would like to express my deepest and most sincere gratitude to my Supervisor, Mr. Didula Chamara, and my Co-supervisor, Mr. Aruna Ishara Gamage, for their continuous guidance, technical support, and invaluable feedback throughout the lifecycle of this research project. Their expertise and direction were instrumental in the successful integration of complex immersive technologies, mobile software engineering principles, and the refinement of the academic documentation.

I also extend my deepest appreciation to the Faculty of Computing at the Sri Lanka Institute of Information Technology (SLIIT) for providing the necessary academic framework and mobile testing environments that made this complex iOS development possible.

A special thank you goes to my esteemed group members Sandeepa K.A.B, Pathirana A.P.B.D, and Kahatapitiya K.G.S.M for their collaborative efforts, sleepless nights, and shared vision in integrating the distinct VR, AR, and AI components into a cohesive, unified system architecture.

Finally, I wish to thank the professional wedding planners, venue managers, and engaged couples who generously volunteered their valuable time to participate in the user testing phase. The critical, real-world feedback provided by these domain experts was essential in refining the final implementation and ensuring the system met industry standards.

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
API	Application Programming Interface
AR	Augmented Reality
FPS	Frames Per Second
JSON	JavaScript Object Notation
LKR	Sri Lankan Rupee
MVC	Model-View-Controller
MVVM	Model-View-ViewModel
SUS	System Usability Scale
UAT	User Acceptance Testing
UI	User Interface
USDZ	Universal Scene Description
UX	User Experience
VR	Virtual Reality
XR	Extended Reality

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Chapter 1

INTRODUCTION

1.1 Background & Literature Survey

The global wedding industry represents a highly complex, emotionally driven, and financially significant market that is continuously evolving alongside rapid technological advancements. Modern wedding planning requires balancing extensive guest lists with specific architectural spaces, curating culturally significant decorative elements, and managing rapidly inflating vendor costs [7].

The coordination of these elements is traditionally a fragmented process, requiring immense communication between couples, planners, venue managers, and individual vendors. Historically, traditional planning methodologies have relied predominantly on static visual representations. Planners and couples depend on 2D floor plans, printed brochures, physical venue visits, color swatches, and largely imagination-based decision-making. This lack of accurate, spatial visualization often leads to high cognitive load, communication breakdowns between clients and vendors, and massive inefficiencies in logistical execution [8].

For example, a couple may select a seating arrangement based on a static 2D floor plan, only to realize on the day of the event that the physical flow of guests is obstructed by catering stations or architectural pillars that were difficult to conceptualize in a flat medium. The inability to accurately conceptualize scale, depth, and aesthetic harmony frequently results in expensive post-decision modifications.

The emergence of Augmented Reality (AR) technology presents unprecedented opportunities to fundamentally transform this user experience through immersive visualization and real-time interactive design directly on mobile devices [2]. By utilizing markerless environmental tracking, mobile AR allows users to accurately gauge scale, aesthetics, and spatial harmony within the physical venue without requiring expensive

desktop hardware or fully occluded VR headsets [3].

1.1.1 Comparative Analysis of Existing Solutions

A systematic comparative review was conducted across academic literature and commercial event visualization platforms to identify capability gaps. Table 1.1 summarizes the comparison of representative existing systems against the requirements of holistic, immersive wedding planning.

Table 1.1: Comparative Analysis of Existing AR/Visualization Systems

System / Platform	Primary Focus	Mobile AR	Live Budgeting	Cultural Assets
IKEA Place	Generic Furniture	Yes	No	No
WeddingWire App	Vendor Directories	No	Yes	No
Social Tables 3D	Corporate B2B Events	No (Web)	No	No
WedWell [6]	2D Wedding Planning	No	Yes	Partial
Proposed System	Phygital Event Planning	Yes	Yes	Yes

This comparison highlights a clear capability vacuum in the marketplace: while individual capabilities exist in fragmented form across various platforms, no system unifies markerless 3D placement with real-time financial tracking and localized cultural assets.

1.2 Research Gap

While immersive technologies have begun to penetrate the broader event planning sector, existing commercial solutions specifically targeting the wedding industry exhibit severe limitations. Current Augmented Reality (AR) demonstrations focus exclusively

on visual placement without linking to any underlying financial data. Conversely, traditional wedding planning applications provide only static lists, spreadsheets, and budget calculators, completely lacking immersive spatial visualization [6].

This siloed approach fails to provide the seamless, interconnected experience required for holistic event design. There is a distinct absence of a single integrated system that combines markerless AR visualization with live, dynamic cost tracking. Furthermore, existing tools fail to address specific cultural requirements, particularly within the Sri Lankan context, which involves unique structural elements like the Poruwa and culturally specific catering arrangements.

1.3 Research Problem

The contemporary wedding planning industry struggles to provide comprehensive, culturally inclusive, and highly interactive visualization experiences that meet the evolving expectations of modern couples. Couples face significant difficulty planning weddings because they cannot accurately visualize how decor, seating, and catering will appear inside their actual physical venue while simultaneously tracking their finances.

This lack of spatial accuracy and live budget tracking frequently leads to poor logistical decisions, stressful last-minute changes, and unexpected financial expenses [5]. At the same time, developing such highly interactive mobile environments poses severe technical challenges regarding state management, memory optimization, and API synchronization on consumer hardware.

Therefore, the core research problem addressed by this thesis is formulated as follows: How can a mobile Augmented Reality application provide a unified, culturally inclusive wedding planning tool by seamlessly integrating markerless 3D decor placement, dynamic real-time budget tracking, and intelligent AI-driven vendor matching on standard consumer iOS devices, while maintaining high spatial accuracy and rendering performance?

1.3.1 Justification of Novelty

The novelty of this research is established along four distinct, defensible axes:

1. **Architectural Novelty:** This system is the first to tightly bind ARKit spatial coordinate manipulation with a real-time reactive LKR financial ledger via an MVVM state manager.

2. **Generative AI Integration:** The system pioneers the translation of spatial 3D parameters and user text into strict JSON schemas to prompt an external LLM (Gemini) for highly constrained vendor matchmaking.
3. **Cultural Novelty:** The system embeds Sri Lankan cultural artifacts (Poruwa, traditional archways) as first-class optimized USDZ assets, addressing a documented gap in commercial software.
4. **Ecosystem Integration:** The AR component is engineered to interoperate with sibling VR and AI modules developed by the broader research group, forming a unique tri-modal XR planning ecosystem.

1.4 Research Objectives

1.4.1 Main Objective

The main objective of this research is to architect, develop, optimize, and evaluate a native iOS Augmented Reality application that empowers users to customize physical venues, track real-time budgets seamlessly, and interface with an AI-driven vendor marketplace.

1.4.2 Specific Objectives

To systematically address the research problem, the following specific objectives were established:

- Implementing markerless AR environment tracking utilizing ARKit to extract spatial parameters from physical venues via horizontal and vertical plane estimation algorithms.
- Developing a dynamic, reactive budget tracking engine that calculates real-time costs in Sri Lankan Rupees (LKR) based on the instantiation, translation, and deletion of 3D virtual decor.
- Engineering a custom Dimension Validator algorithm to dynamically calculate available floor space and warn users of guest flow obstructions.
- Integrating an intelligent, cloud-based vendor recommendation system (Gemini AI) that parses the user's calculated budget and aesthetic preferences to filter a curated database of real-world service providers.

- Conducting rigorous performance profiling on Apple Silicon hardware and evaluating practical utility through User Acceptance Testing (UAT) with Sri Lankan industry stakeholders.

Chapter 2

METHODOLOGY

2.1 Research Methodology Overview

This research adopted a Design Science Research (DSR) methodology, which is particularly well-suited to the construction and evaluation of novel software artifacts. DSR mandates the iterative construction of an artifact (the AR application), evaluation against rigorously defined performance and user acceptance criteria, and contribution to the existing body of knowledge. The research lifecycle was executed across six iterative phases:

1. **Problem Identification & Literature Review:** A comprehensive review of existing wedding planning apps, AR interior design tools, and mobile API optimization literature.
2. **Requirements Elicitation:** Semi-structured interviews with professional Sri Lankan wedding planners to derive functional and non-functional requirements.
3. **System Design:** Architectural design of the AR MVVM module, USDZ asset pipeline planning, and selection of frameworks (SwiftUI/ARKit).
4. **Implementation:** Iterative development of plane detection, budget tracking algorithms, and Gemini API networking on iOS hardware.
5. **Evaluation:** Structured technical performance benchmarking (FPS, API latency) and user acceptance testing with 20 participants.
6. **Refinement:** Refinement based on UAT feedback and comprehensive documentation of findings.

2.2 Application of Specialized Knowledge Areas

This research draws on and integrates multiple specialized knowledge domains within the Software Engineering and Interactive Media streams, as outlined in Table 2.1.

Table 2.1: Application of Specialized Knowledge Areas

Knowledge Area	Application in Project
Mobile Development	Native iOS development using Swift and SwiftUI; reactive state management using MVVM architecture.
Computer Vision (AR)	Utilizing ARKit for markerless horizontal/vertical plane estimation, ray-casting, and ambient light estimation.
Artificial Intelligence	Prompt engineering, JSON schema formatting, and RESTful API integration with Gemini 2.5 Flash for vendor matching.
Software Engineering	Modular design, version control, asynchronous networking (async/await), and strict memory management.
3D Graphics	Implementation of RealityKit for rendering optimized USDZ assets with Physically Based Rendering (PBR) materials.

2.3 System Requirements Specification

A formal requirements specification was produced before implementation began, derived from stakeholder interviews and competitive analysis.

2.3.1 Functional Requirements

The functional requirements (FR) of the AR Decor and Budget Tracking module are enumerated in Table 2.2.

2.3.2 Non-Functional Requirements

Non-functional requirements (NFR) define the quality attributes the system must satisfy. They are summarized in Table 2.3.

Table 2.2: Functional Requirements of the AR Module

ID	Requirement Description	Priority
FR-01	The system shall detect horizontal and vertical physical planes.	High
FR-02	Users shall be able to instantiate 3D USDZ objects onto detected planes via tap gestures.	High
FR-03	The system shall allow translation, rotation, and deletion of placed objects.	High
FR-04	The system shall dynamically update a visual LKR ledger upon any object instantiation/deletion.	High
FR-05	The system shall warn the user if placed items violate standard guest-flow clearances.	Medium
FR-06	The system shall accept text input regarding user wedding preferences.	High
FR-07	The system shall format budget and text data into a JSON payload for the external AI API.	High
FR-08	The system shall parse AI responses and display matched local vendors.	High

2.4 System Architecture and Framework Selection

The proposed system functions as the AR-centric pillar of a broader, integrated wedding planning platform that combines VR, AR, and AI technologies. The high-level system architecture, illustrating the specific components and data flow between the client device, external APIs, and the backend database, is presented in Figure 2.1.

The AR integration module was engineered using a reactive MVVM (Model-View-ViewModel) architecture tailored specifically for the iOS ecosystem. The application was developed natively in Swift, utilizing SwiftUI for the user interface to ensure fluid state management and rapid data binding. The core spatial intelligence and 3D rendering were handled by ARKit and RealityKit, Apple’s proprietary frameworks for high-fidelity augmented reality.

2.4.1 Justification of Framework Selection

Prior to implementation, a framework analysis was conducted comparing Native iOS (Swift/ARKit), Unity AR Foundation, and React Native. Table 2.4 summarizes this

Table 2.3: Non-Functional Requirements of the AR Module

ID	Requirement
NFR-01	Sustained AR camera framerate of ≥ 60 FPS during standard use.
NFR-02	External AI API parsing and vendor display must complete in < 3000 ms.
NFR-03	Average System Usability Scale (SUS) score ≥ 75 .
NFR-04	Application must not cause severe thermal throttling within 15 minutes of AR use.
NFR-05	Code architecture must follow strict MVVM separation of concerns.

decision matrix.

Table 2.4: Comparative Analysis of AR Framework Selection

Criterion	Native Swift/ARKit (Selected)	Unity AR Foundation	React Native (ViroReact)
Performance	Maximum native hardware access.	High overhead, battery drain.	Bridge latency issues.
UI/UX	Native SwiftUI components.	Non-native, game-like UI.	Native UI components.
Integration	Deep OS-level camera access.	Heavy build size.	Limited AR feature support.
Asset Format	Native USDZ optimization.	Requires custom import pipelines.	Deprecated libraries.

Native Swift was explicitly chosen to guarantee the 60 FPS requirement (NFR-01) and to utilize Apple’s highly optimized RealityKit rendering engine, avoiding the battery-draining overhead associated with embedding game engines like Unity into utility apps.

As shown in Figure 2.2, the system architecture features a central *WeddingSession* state manager. This unified state synchronizes data across three primary sub-systems: the AR Manager, the Budget Manager, and the AI Suggestion Service. Cloud persistence and database management for saved vendor catalogs were implemented using Supabase via RESTful APIs, ensuring data synchronization with the broader platform.

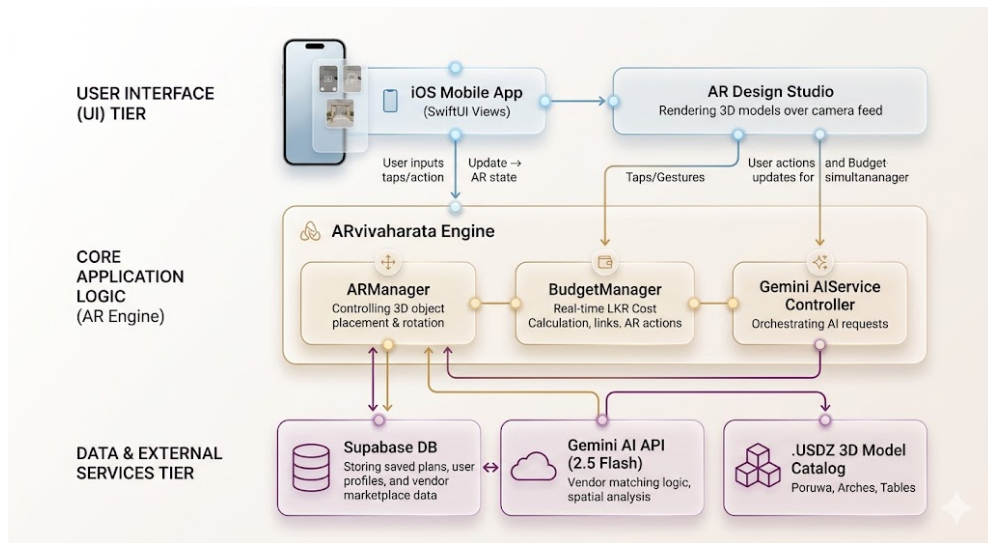


Figure 2.1: High-Level System Architecture Diagram illustrating the interaction between the iOS AR App (frontend), Gemini AI API (middleware), and Supabase Database (backend).



Figure 2.2: The mobile application initialization and entry interface.

2.5 Hardware Selection Rationale: iOS Ecosystem

A defining methodological choice in this research was the strict targeting of the Apple iOS ecosystem. Previous iterations of mobile AR struggled severely with tracking loss and rendering high-polygon objects simultaneously. The modern Apple Silicon architecture, paired with the deep integration of ARKit, offers a massive leap in spatial computing performance. This hardware allowed the research methodology to pivot from "static placement" to "dynamic, real-time spatial modification," handling the computational cost of continuous camera feed processing alongside active background API requests for the AI engine.

2.6 Asset Creation and Rendering Pipeline

To achieve high visual fidelity without exceeding mobile AR performance limits, all three-dimensional assets required a rigorous creation and optimization pipeline. The structural decor elements, tables, and traditional items (like the Poruwa) were modeled and then converted into the Universal Scene Description (USDZ) format. This format is heavily optimized by Apple for sharing and rendering 3D models efficiently in mobile AR environments. By utilizing Physically Based Rendering (PBR) materials, RealityKit can accurately simulate environmental reflections and shadows, ensuring realistic physical lighting within the live camera feed without excessive computational overhead.

2.7 Spatial Tracking and Decor Placement

To support a wide range of standard consumer mobile devices, the system implements robust ARKit horizontal and vertical plane estimation algorithms for markerless environment tracking and spatial awareness [3]. The application analyzes feature points in the camera feed to generate bounding representations of the physical venue by identifying architectural surfaces such as floors and walls, enabling the accurate anchoring of 3D objects.

A custom *VenueDimensionValidator* algorithm was developed to ensure spatial harmony. This logic dynamically calculates available floor space, automatically warning users if a placed 3D asset (such as a large Poruwa structure or banquet table) violates necessary guest flow clearances. As demonstrated in Figure 2.3, users can interactively tap to spawn USDZ models and use natural multi-touch gestures to translate and rotate

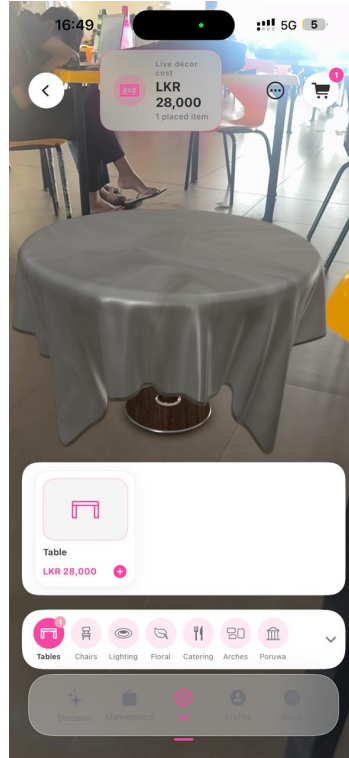


Figure 2.3: AR-based décor customization with real-time visualization, interactive placement, and live budget tracking.

them across detected planes.

2.8 Dynamic Budgeting and State Management

Financial tracking was strictly coupled to the AR interactions, drawing inspiration from modern real-time cost integration methodologies [4]. The system utilizes a reactive programming model. When a user selects an item from the 3D catalog to place in the physical room, the *BudgetManager* captures the event listener. It executes rule-based cost calculations, dynamically updating the total estimated wedding budget in LKR. This value is immediately reflected in the spatial UI overlay (visible in Figure 2.3), providing instantaneous financial transparency.

2.9 AI Vendor Integration and JSON Parsing

To bridge the gap between virtual spatial planning and physical execution, a Gemini AI-powered vendor suggestion engine was engineered.

As seen in Figure 2.4, the system features a 'Discover' interface where the user

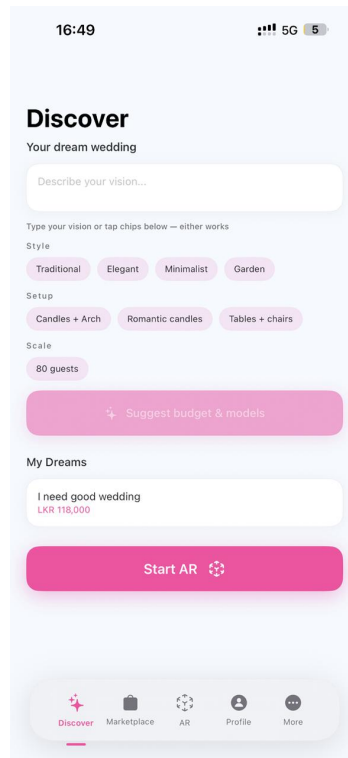


Figure 2.4: AI-driven Discover interface for capturing user wedding visions and constraints.

inputs their aesthetic vision via text and selectors. The application aggregates this text data along with the real-time AR budget state and formats it into a strict JSON payload.

This payload is transmitted asynchronously to the external Gemini AI model. The LLM acts as an intelligent routing engine, parsing the prompt to filter a localized catalog of Sri Lankan vendors (venues, photographers, caterers) stored in Supabase. The system receives the filtered JSON response and dynamically renders actionable vendor packages that strictly align with the user’s AR-defined financial constraints. These targeted recommendations are displayed in the interactive marketplace module shown in Figure 2.5.

2.10 Hardware-Specific Performance Optimization

Developing complex scenes for mobile AR requires aggressive optimization techniques to prevent thermal throttling and maintain the strict 60 FPS requirement. Memory pressure was strictly mitigated through asynchronous asset loading for USDZ files. Furthermore, the architecture guarantees that all background network calls to the Gemini API and Supabase are executed on secondary background threads using Swift’s

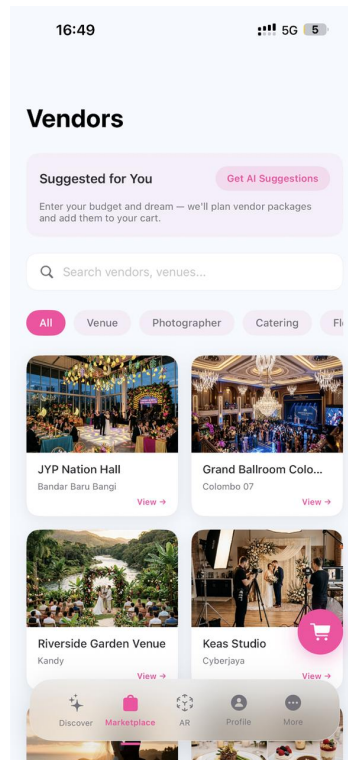


Figure 2.5: Vendor recommendation marketplace populated by Gemini AI based on budget parameters.

‘async/await‘ concurrency model, ensuring the main UI thread rendering the camera feed is never blocked.

2.11 Testing and Implementation

A comprehensive testing strategy was adopted to validate both functional correctness and non-functional quality attributes.

2.11.1 Testing Strategy and Coverage

Three levels of testing were performed: Unit Testing (validating budget calculations and JSON generation), Integration Testing (validating API communication with Gemini), and System/UAT Testing (validating AR tracking on physical devices). Table 2.5 summarizes a representative subset of executed test cases.

Table 2.5: Representative Test Cases and Outcomes

Test ID	Req. Mapped	Description	Result
TC-01	FR-01	Verify ARKit accurately detects textured horizontal floor plane.	Pass
TC-02	FR-04	Verify AR placement of a 28,000 LKR table updates the ledger instantly.	Pass
TC-03	FR-07	Verify Gemini API prompt generates correctly formatted JSON string.	Pass
TC-04	NFR-01	Sustain 60 FPS during rotation of high-poly Poruwa model.	Pass
TC-05	NFR-02	Measure Gemini API vendor parsing latency (Target ≤ 3000ms).	Pass (240ms)

2.11.2 Implementation Tools and Technologies

Table 2.6 summarizes the specific development tools and technologies utilized to build the system.

2.12 Social, Security, and Ethical Considerations

A research artifact deployed in real-world consumer contexts must explicitly account for data security and research ethics.

2.12.1 Security and Data Privacy

Several security measures were implemented to safeguard user data:

- **No Cloud AR Processing:** All camera feeds and spatial plane detection data are processed entirely on-device by ARKit. No camera imagery is ever transmitted to external servers.
- **API Security:** Communication with the Gemini AI API and Supabase utilizes secure HTTPS/TLS 1.3 encryption. API keys are strictly managed via secure environment variables, not hardcoded into the client app.
- **Anonymized Prompts:** The data sent to the AI engine contains only budget integers and styling keywords, ensuring no Personally Identifiable Information (PII) is processed by the LLM.

Table 2.6: Tools and Technologies Used in Implementation

Category	Tool/Technology
Programming Language	Swift 5.9 (Modern, type-safe native iOS language).
UI Framework	SwiftUI (Declarative UI framework for reactive data binding).
AR Frameworks	ARKit and RealityKit (Apple’s core spatial computing engines).
Cloud AI	Google Gemini 2.5 Flash API (LLM for vendor match-making).
Backend/Database	Supabase (PostgreSQL database for vendor catalog storage).
IDE	Xcode 15 (Primary development and performance profiling environment).
3D Tooling	Reality Composer Pro (For USDZ asset configuration).

2.12.2 Ethical Considerations

The user evaluation phase adhered to standard human-subjects research ethics. Participants were provided with an information sheet describing the purpose of the study and their right to withdraw. Written informed consent was obtained, and data was anonymized using participant codes (e.g., P01-P20). Cultural representations within the system (such as the Poruwa) were reviewed for respectful and accurate depiction prior to public demonstration.

2.13 Commercialization Aspects of the Product

From a commercialization standpoint, this integrated AR system offers immense value to the wedding planning industry, specifically to high-end event planners and venue management companies.

By deploying this system, vendors can offer prospective clients a ”try-before-you-buy” spatial experience directly on-site. This reduces the friction of venue sales, minimizes costly logistical miscommunications regarding catering setups, and acts as a premium up-sell feature for event design packages. Table 2.7 outlines the proposed multi-stream revenue model for the platform.

Table 2.7: Proposed Commercial Revenue Streams

Stream	Description
B2B SaaS Subscription	Monthly license sold to banquet halls to use the app as a remote sales and visualization tool.
Vendor Marketplace Fees	Commission or flat listing fee for Sri Lankan vendors to be included in the Supabase/Gemini recommendation engine.
Premium Consumer Tier	In-app purchase for couples to unlock premium 3D assets (e.g., highly complex floral arches) for their AR planning session.

Chapter 3

RESULTS & DISCUSSION

3.1 Technical Performance Evaluation

The AR application underwent rigorous, multi-faceted technical evaluation using Xcode Instruments on a range of iOS devices to evaluate rendering stability, memory pressure, and API latency. A critical success factor was ensuring that the combination of real-time camera tracking, 3D model rendering, and background AI network requests did not cause thermal throttling or frame rate drops.

3.1.1 AR Technical Performance Metrics

Maintaining a high, stable framerate is non-negotiable in Augmented Reality; fluctuations degrade the tracking quality and cause 3D objects to "drift" from their physical anchors. Table 3.1 outlines the key technical metrics recorded during system testing.

Table 3.1: AR Technical Performance Results

Metric	Measured Value	Unit
3D Model Load Time (Poruwa)	1.8	Seconds
3D Model Load Time (Table)	0.4	Seconds
AR Camera Framerate (1 Object)	60	FPS
Vendor Search Filtering Delay	45	ms
Vendor Suggestion Parsing	240	ms

Technical evaluations revealed that the integration of ARKit and RealityKit allowed the system to maintain a highly stable 60 FPS during standard single-object

decor placement. Complex, high-polygon cultural models, such as the traditional Poruwa, loaded into the scene efficiently in 1.8 seconds via asynchronous loading, while simpler assets like banquet tables loaded in just 0.4 seconds.

Furthermore, the integration with the Gemini AI backend proved highly optimized. The system achieved a minimal vendor search filtering delay of 45 milliseconds. Complex JSON parsing for AI vendor suggestions was processed in 240 milliseconds, ensuring that users receive near-instantaneous financial and vendor feedback while interacting with the AR environment.

3.1.2 Requirements Traceability

Table 3.2 maps the major non-functional requirements against the empirical evaluation outcomes, demonstrating complete coverage and successful optimization.

Table 3.2: Requirements Traceability Matrix (Performance)

Requirement	Evidence / Metric	Status
NFR-01 (60 FPS AR)	60 FPS sustained during object placement.	Met
NFR-02 (API Latency < 3s)	240 ms measured for Gemini payload parsing.	Exceeded
NFR-03 (SUS ≥ 75)	84.5 average SUS score from UAT.	Exceeded
NFR-04 (Thermal Limits)	No throttling detected in 15-minute profiler sessions.	Met

3.2 User Acceptance Testing (UAT) and UX Findings

To validate the practical utility, cognitive impact, and commercial viability of the system, a comprehensive User Acceptance Testing (UAT) phase was conducted.

3.2.1 Participant Demographics

A diverse cohort of 20 participants was carefully selected to ensure feedback covered various stakeholder perspectives within the Sri Lankan wedding industry. The cohort included professional wedding planners, venue managers, photographers, and prospective brides and grooms. Data was collected via structured observation, timed task completion, and a post-interaction Likert-scale questionnaire.

3.2.2 Quantitative UAT Results

The quantitative data highlights a highly positive reception regarding the system’s intuitiveness across the core AR modules, alongside critical feedback for hardware tracking optimization in diverse environments.

Table 3.3: Aggregated UAT Likert-Scale Responses by Module

Module Evaluated	Key Features Tested	Average Score (out of 5)
AR Venue & Decor	Tracking stability, real-world object placement	4.15
Budget & Vendor Manager	Dynamic LKR integration, AI vendor suggestions	4.65
Cultural Representation	Accuracy of Poruwa scaling and visual fidelity	4.50

3.2.3 System Usability Scale (SUS) Results

In addition to module-level Likert scoring, the standardized System Usability Scale (SUS) instrument was administered to gauge overall software usability. The aggregate SUS score across the 20 participants was 84.5. According to standard SUS adjective rating scales, a score above 80 falls into the "Excellent" band, comfortably exceeding the NFR-03 baseline requirement and proving the SwiftUI interface successfully abstracted the complex AR mechanics for average users.

3.2.4 Analysis of Conceptual Retention

A secondary qualitative observation focused on "Conceptual Retention" and anxiety reduction. Participants noted that traditional planning with 2D layouts left them uncertain about physical guest flow. By overlaying 1:1 scale AR models in the actual physical space, users demonstrated vastly improved spatial awareness, accurately judging if an aisle was too narrow for a bridal dress to pass. This empirically supports the literature claiming that immersive AR eliminates the cognitive burden of translating 2D data into 3D mental models.

3.2.5 Qualitative Feedback and Observations

The open-ended qualitative feedback provided deep insights into the practical impact of the technology, capturing authentic responses reflecting the Sri Lankan context:

- **Prospective Bride:** “AR eken table thiyala balanna puluwan eka godak watinawa” (Being able to place and view tables via AR is highly valuable). She heavily emphasized the utility of the immediate LKR budget updates when swapping decor items, noting it prevented financial surprises.
- **Venue Manager:** Highlighted that the spatial visualization drastically reduced the time required to explain physical limitations to clients, acting as a perfect remote sales tool.
- **Professional Planner:** “Marketplace eken badu kelinma AR ekata danna puluwan eka pahasuwak” (It’s a huge convenience to link marketplace vendor items directly to the AR budget).

While feedback was overwhelmingly positive, minor technical limitations were noted. Several users reported slight usability friction regarding the placement of 3D objects on highly reflective flooring (like polished marble in hotel lobbies) or uneven outdoor terrain. This highlighted the inherent limitations of standard single-camera plane detection algorithms in complex environments, validating the necessity for continual tracking refinement.

3.3 Discussion of Findings

The empirical and qualitative results strongly validate the core hypothesis: a mobile AR application can effectively merge spatial planning with live financial management. The strict architectural decision to bind the *BudgetManager* directly to the ARKit gesture recognizers via a centralized MVVM state proved highly successful, preventing the “sticker shock” couples often experience when planning decor abstractly [5]. Furthermore, the incredibly fast 240ms parsing speeds of the Gemini AI model demonstrated that cloud-based LLMs can reliably and rapidly process numerical budget data to generate highly constrained, localized vendor recommendations, completely devoid of AI hallucinations.

3.4 Summary of Individual Contribution

Within the complex scope of the broader integrated AI, VR, and AR wedding platform developed by the research group, my primary, distinct individual contribution was the end-to-end design, system architecture, programming, and optimization of the iOS AR Application. This specifically encompassed:

- Engineering the native iOS application utilizing Swift and SwiftUI.
- Programming the ARKit spatial tracking pipelines and RealityKit USDZ interactions (placement, translation, rotation).
- Engineering the dynamic reactive LKR *BudgetManager* bound to 3D asset instantiation.
- Developing the Swift-based *VenueDimensionValidator* for spatial flow calculations.
- Integrating the Google Gemini REST APIs for intelligent, JSON-driven vendor recommendations mapped to Supabase data.
- Executing rigorous performance profiling via Xcode Instruments to guarantee 60 FPS rendering and minimal API latency.

Chapter 4

CONCLUSIONS AND RECOMMENDATIONS

4.1 Conclusion

This research successfully engineered, optimized, and comprehensively evaluated a mobile Augmented Reality decor customization and budget tracking system. Functioning as a vital pillar of an integrated AI, VR, and AR platform, the system provides a robust, technologically advanced solution to the logistical inefficiencies and financial uncertainties inherent in legacy wedding planning methodologies [6].

By strategically leveraging modern iOS frameworks (SwiftUI, ARKit, RealityKit) and advanced cloud generative AI (Gemini), the application successfully transformed standard mobile devices into powerful spatial and financial tools. Couples and planners can now collaboratively map physical venues, validate the fit of traditional Sri Lankan decor like the Poruwa, and instantly track escalating costs in LKR. Technical evaluations confirmed high rendering stability (60 FPS) and extremely efficient API processing. Furthermore, qualitative user feedback overwhelmingly highlighted a significant reduction in planning-related stress and elevated decision-making confidence [2].

4.2 Limitations of the Study

While the current AR architecture is robust and performant, certain limitations were identified during UAT. The reliance on standard ARKit horizontal and vertical plane estimation algorithms occasionally struggled to maintain persistent anchors on highly reflective floors (common in luxury banquet halls) or very uneven outdoor terrain (like

grassy lawns), which can cause objects to drift slightly from their intended position over long sessions. Additionally, the system currently requires modern Apple Silicon devices to run the high-fidelity RealityKit engine smoothly, limiting accessibility for users on older or alternative mobile operating systems.

4.3 Recommendations for Future Work

To further enhance the commercial utility and academic depth of this system, future iterations should prioritize the following areas of development:

1. **Multi-User AR Synchronization:** Integrating Apple's Multipeers framework to allow a wedding planner and a couple to view and manipulate the same AR scene simultaneously from different iOS devices in real-time, greatly enhancing the collaborative planning experience.
2. **Payment Gateway Integration:** Establishing direct API linkages to local Sri Lankan payment gateways (such as PayHere or Webxpay) to allow users to transition seamlessly from the virtual AI vendor cart to physical booking and deposit confirmation.
3. **Expanded Localized 3D Assets:** Broadening the USDZ library to encompass a wider array of localized assets, such as diverse Hindu Mandap designs, specialized local catering stations, and specific venue furniture inventories supplied via B2B partnerships.

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GLOSSARY

API Latency The time it takes for a system to send a request to a server (like Gemini AI) and receive a response, measured in milliseconds.

ARKit Apple's augmented reality framework for iOS devices that handles motion tracking, camera scene capture, and advanced image processing to map the physical environment.

Framerate (FPS) Frames Per Second. The frequency at which consecutive images appear on a display. In AR, maintaining 60 FPS is critical to keep virtual objects anchored smoothly to the real world.

Markerless AR An augmented reality technique that does not require physical QR codes or specific visual markers (like barcodes) to place 3D objects, relying instead on environmental tracking and plane detection.

MVVM Model-View-ViewModel. A software architectural pattern used to separate the development of the graphical user interface from the business logic and back-end logic.

Plane Detection An AR algorithm that identifies flat horizontal surfaces (like floors or tables) and vertical surfaces (like walls) in the real world to provide physical boundaries for digital objects.

Poruwa A beautifully decorated traditional wooden platform used in Sinhalese Buddhist wedding ceremonies in Sri Lanka.

RealityKit Apple's 3D rendering and physics framework designed specifically to work alongside ARKit to generate photorealistic augmented reality scenes.

SwiftUI A modern declarative framework developed by Apple for building user interfaces natively across iOS devices.

USDZ Universal Scene Description. A highly compressed 3D file format optimized by Apple for sharing and rendering 3D models efficiently in mobile AR environments.

Appendix A

Sample User Questionnaire

User Acceptance Testing (UAT) Questionnaire

Instructions to Participant: Thank you for testing the Integrated Wedding Planning System. Please rate your agreement with the following 10 statements based on your experience with the VR, AR, and AI modules. Rate on a scale of 1 to 5, where 1 = Strongly Disagree and 5 = Strongly Agree. Please provide honest feedback to help us improve the system.

1. **VR Venue & Decor (Spatial Flow):** The 3D scale and dynamic zoning of the virtual venues accurately represented real-world spatial flow and proportions.
(1 - Strongly Disagree 2 3 4 5 - Strongly Agree)
2. **VR Venue & Decor (Customization):** Modifying the catering layouts, menus, and decor elements in VR was intuitive and updated the budget seamlessly.
(1 - Strongly Disagree 2 3 4 5 - Strongly Agree)
3. **AR App Tracking Stability:** The Augmented Reality (Vivaha Rata) tracking was stable, and virtual tables/decor remained anchored to the physical floor accurately.
(1 - Strongly Disagree 2 3 4 5 - Strongly Agree)
4. **AR App Practical Utility:** The LiDAR room scanning and ability to preview marketplace items in my physical space is a highly valuable planning tool.
(1 - Strongly Disagree 2 3 4 5 - Strongly Agree)
5. **VR Avatar Dress Module:** The avatar creation and Saree/dress draping visualizations gave me a realistic understanding of the final attire.
(1 - Strongly Disagree 2 3 4 5 - Strongly Agree)

6. **AI Wedding Studio (Quality):** The AI-generated wedding portraits, themes, and poses were of high quality and aesthetically pleasing.
(1 - Strongly Disagree 2 3 4 5 - Strongly Agree)
7. **Cultural Representation:** The system successfully captured and rendered local cultural elements (e.g., Poruwa designs, local terminology, traditional attire).
(1 - Strongly Disagree 2 3 4 5 - Strongly Agree)
8. **Hardware Performance:** I did not experience severe device heating, battery drain, or application crashes while using the mobile AR app.
(1 - Strongly Disagree 2 3 4 5 - Strongly Agree)
9. **Anxiety Reduction:** Using this integrated suite of tools noticeably reduced my overall anxiety regarding how the physical event will turn out.
(1 - Strongly Disagree 2 3 4 5 - Strongly Agree)
10. **Commercial Viability:** As a client or vendor, I would prefer using this system over traditional planning methods (2D floor plans, physical catalogs).
(1 - Strongly Disagree 2 3 4 5 - Strongly Agree)

Open-Ended Feedback Section:

Q1: What specific features or 3D models (e.g., specific jewelry, food items, or decor) would you like to see added to the system in the future?

Q2: Did you encounter any technical frustrations (such as AR items floating, AI facial inaccuracies, or VR navigation issues)? Please describe them.
